SCALER

UNSUPERVISED LEARNING ALGORITHMS

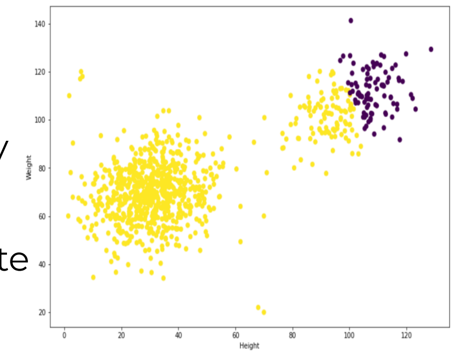
1. Clustering

It is a task of **grouping a set of points/observations** (without any target variable) so that points in the same group are more similar to each other than those in other groups.

	Distance-Based Clustering		Density-Based Clustering	Distribution-Based Clustering
Name	K-Means	Hierarchical	DBSCAN (Density-based spatial clustering application with noise)	Gaussian Mixture Models (GMM)
Main Idea	Starting with **K random centroids**, we assign points to each of them to form **K-clusters**	Starting from each point as each cluster, we **group similar points until there is only one cluster**. The **ideal K** is obtained using a **dendrogram**	Idea is to **classify points as either core point, border point** or a **noise point** based on **how densely** a point is **surrounded** by other points.	Given that **data follows gaussian distribution**, we identify **mean and variance** that best represents the shape of the clusters.
Online	K - number of clusters	**Linkage** Function	Type of **distance metric** chosen for calculating distances between points.	**μ(mean)** and **σ(variance)**
Hyper-parameters	Yes	No	No	Yes
Advantages	Simplest clustering algorithm.	No need to decide K before clustering	Good ability to **find arbitrarily shaped clusters** and **clusters with noise.** No need to decide K	Provides more intuitive results for making decisions. Provides a **lot more elliptical clusters.**
Dis-advantages	• Random initialization problem • Fails for varying size or density, and non-globular shapes. • Need to define K.	• Very sensitive towards the choice of linkage functions. • It is an offline model.	• Can't handle high dimensional data. • DBSCAN struggles with clusters of similar density.	• It assumes normal distribution of features. • Clusters are assumed to be elliptical. • Need to specify the number of clusters.

2. Anomaly Detection

- Anomaly is synonymous with an outlier. Anomaly means something which is not a part of normal behavior.
- Novelty means something unique, or something that you haven't seen before(novel).

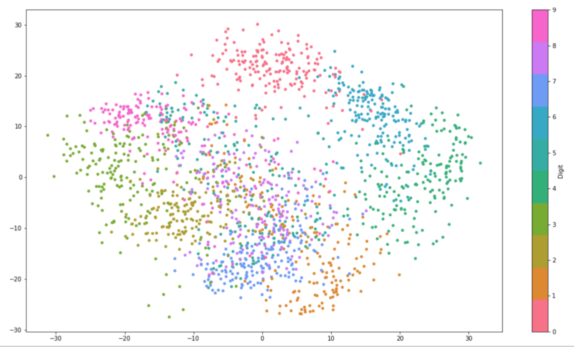
	Elliptic Envelope	Isolation Forest	Local Outlier Factor (LOF)
Main Idea	Mark the points as outliers which are **very far away from the centroid** of the ellipse.	We **randomly make splits** in the data and make **trees out of it** until there is only a **single point in the leaf node**. On an average, outliers have lower depth and inliers have more depth in the random trees.	The core idea behind LOF is to **compare the density of a point** with its neighbors' density. If the density of a point is less than the density of its neighbors, we flag that point as an outlier. $LOF_k(A) = \frac{\sum_{B \in N_k(A)} lrd_k(B)}{\lvert N_k(A)\rvert . lrd_k(A)}$
Drawbacks	• Does not work on non-unimodal data. • It is specifically for multivariate Gaussians  • Data is assumed to follow unimodal and multivariate gaussian.	• Biased towards axis-parallel splits	• Need to find optimal K • Need to tune threshold. • Bad performance on high dimensional data • High Time Complexity

3. High Dimension Visualization

Dimensionality reduction techniques help to **convert high dimensional data to fewer dimensions** preserving the information of our feature columns.

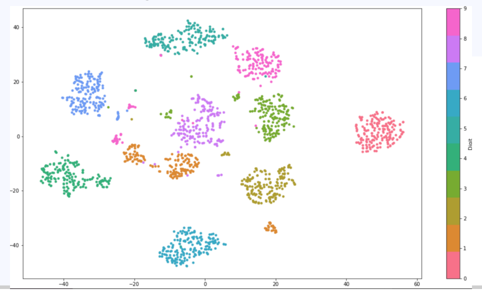
A. Principal Component Analysis (PCA)

- The main idea of PCA is to find the **best value of vector u**, which is the **direction of maximum variance** (or maximum information) and along which we should rotate our existing coordinates.
- The eigenvector associated with the largest eigenvalue indicates the direction in which the data has the most variance.



B. t-SNE (t-Distributed Stochastic Neighbor Embedding)

- t-SNE tries to **create an embedding that preserves the neighborhood** using some probabilistic methods when the datapoint in a higher dimensional space is projected into a lower dimensional space.
- We compute $\mathbf{P_{ij}}$ for d-dimensions and $\mathbf{Q_{ij}}$ for d'-dimensions where d>d.



$$p_{ij} = \frac{exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq l} exp(-\|x_k - x_l\|^2 / 2\sigma_i^2)}$$

- We define $\mathbf{q_{ij}}$ with the same formulation as $\mathbf{p_{ij}}$ since every $\mathbf{x_i}$ and $\mathbf{x_j}$ would have corresponding $\mathbf{y_i}$ and $\mathbf{y_j}$ in d' dimensional space.

$$q_{ij} = \frac{exp(-\|y_i - y_j\|^2)}{\sum_{k \neq l} exp(-\|y_k - y_l\|^2)}$$

- For a useful d' transformation we need $\mathbf{p_{ij} \approx q_{ij}}$, we use **KL-divergence** that defines a **loss function** which measures **the dissimilarity between two distributions.**
- t-distributions work better than gaussian distributions with SNE because:
  - t-distributions with a **degree of freedom equal to 1** have a **longer tail** than gaussian distributions
- Gaussian distribution **falls exponentially** while t-distributions **sort of inversely.**

4. Recommender System

Recommender systems are **designed to recommend things to the user based on many factors.** These systems predict the most likely product the users are most likely to purchase and are interested in.

4.1 Market-Basket Analysis

Market basket analysis is used to analyze the **combination of products that have been bought together.**

Association Rules

The **IF component** of an association rule is known as the **antecedent**. The **THEN component** is known as the **consequent.**

Support:

$$Support(x) = \frac{Number\ of\ transactions\ with\ x}{Total\ number\ of\ transactions}$$

Confidence:

$$Confidence(X \to Y) = \frac{Number\ of\ transactions\ with\ X\ and\ Y}{Number\ of\ transactions\ with\ X}$$

Lift:

$$Lift(X \to Y) = \frac{Confidence(X \to Y)}{Support(Y)} = \frac{Support(X \cap Y)}{Support(X) * Support(Y)}$$

Pros	Cons
• It is the most simple and easy to understand and implement. • It is used to **calculate large item sets.**	• It is computationally expensive. • **Complexity grows exponentially.** • Cold start problem.

4.2 Content-Based Recommender System

Recommends items with **similar content (e.g. metadata, description, topics)** to the items the user has liked in the past.

Pros	Cons
• **No cold start problem** • No need for usage data • **No popularity bias,** can recommend items with rare features • Can **capture user content** features to provide recommendations	• It **always recommends items related to the same categories,** and never recommend anything from other categories. • Requires a lot of domain knowledge.

Recommender System

4.3 Collaborative Filtering System

This system looks for **patterns in user activity** to produce user-specific recommendations.

- ***User-Based:*** This is a form of collaborative filtering for recommender systems based on the **similarity between the users** is calculated.
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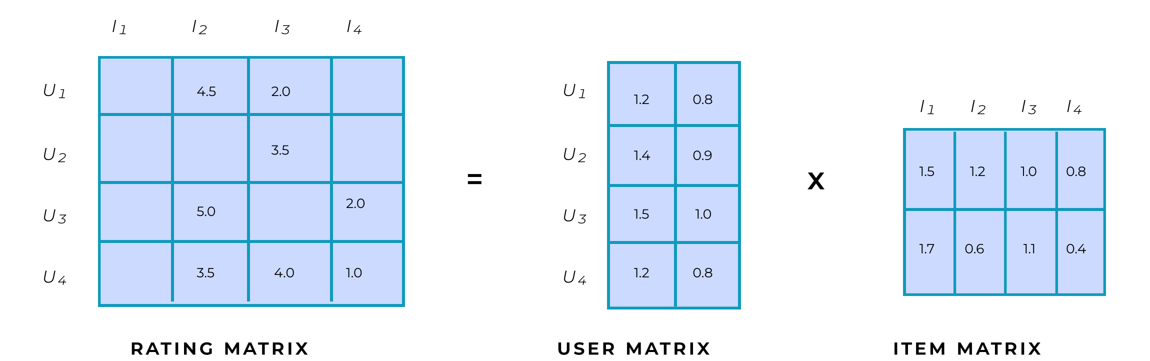
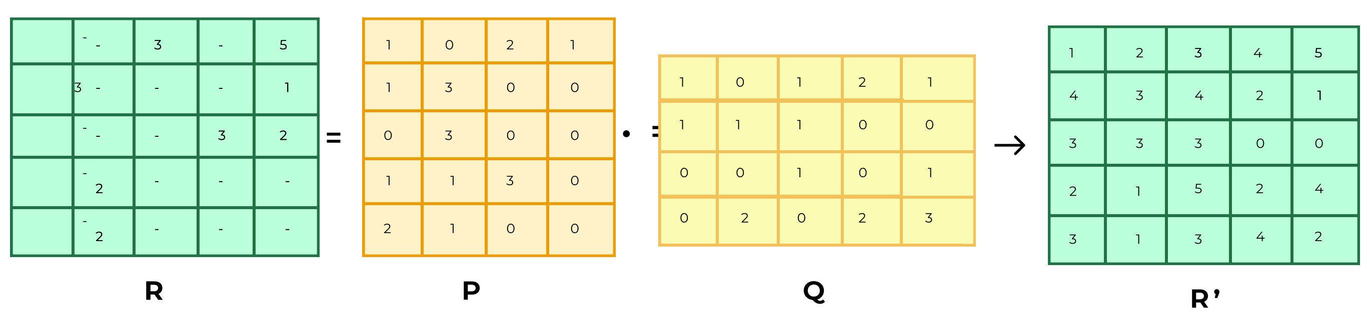
- ***Item-Based:*** This is a form of collaborative filtering for recommender systems based on the similarity between the items is calculated.
- ***Model-based***: This system is based on the similarity between the users and items is calculated. Data contains a set of users and items and ratings/reactions in the form of a user-item interaction matrix.
- ***Matrix Factorization:*** Matrix factorization is a way to generate latent features when multiplying two different kinds of entities. Collaborative filtering is the application of matrix factorization to identify the relationship between items and user entities.



Predicted Ratings: $r'_{ui} = p_u^T q_i$

Latent Feature : k = 4



PROS	CONS
• Minimal domain knowledge required • The system doesn't need contextual features. • Serendipity	• Cold start problem • Computationally expensive. • It's a bit difficult to recommend items to users with unique tastes.

5. Time Series Analysis

- Time Series forecasting is method of making **predictions** based on **historical time-stamped data.**

Trend : It is a linear **increasing or decreasing** behavior of the series over a long period that **does not repeat.**

Seasonality: Seasonality in time-series data refers to a **pattern that occurs at a regular interval.**

Moving Average: The approach of taking an **average of the last k data points** in our series and use it to guess the next point at t=k is Moving Average.

$$y_t = \frac{1}{k}\sum_{i=1}^{k} y_{t-k}$$

5.1 Time Series Decomposition

Additive $y(t) = b(t) + s(t) + e(t)$

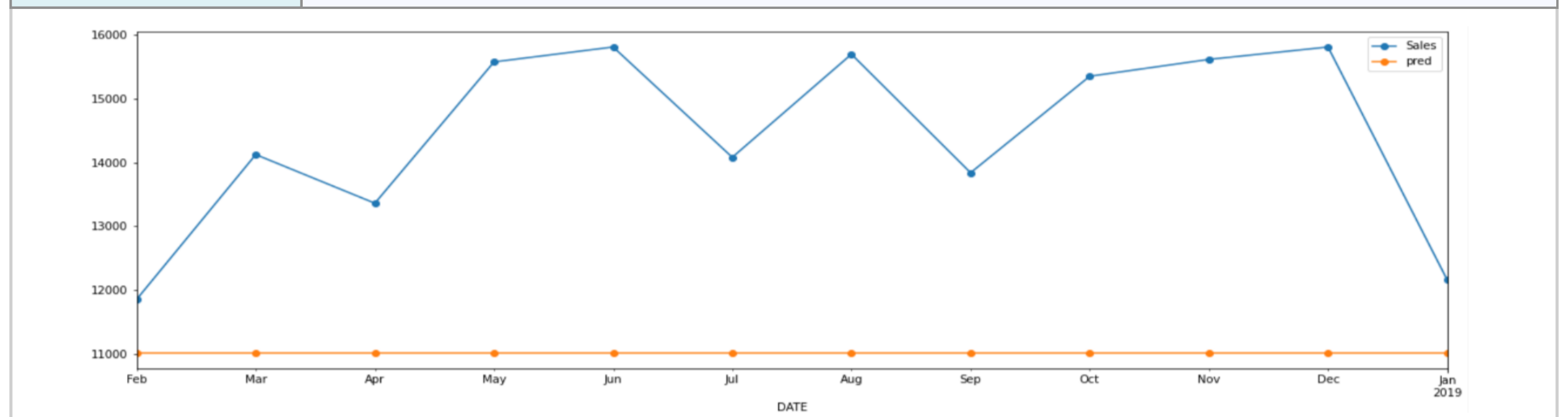
Multiplicative $y(t) = b(t) * s(t) * e(t)$

$$e(t) = \frac{y(t)}{b(t)*s(t)}$$

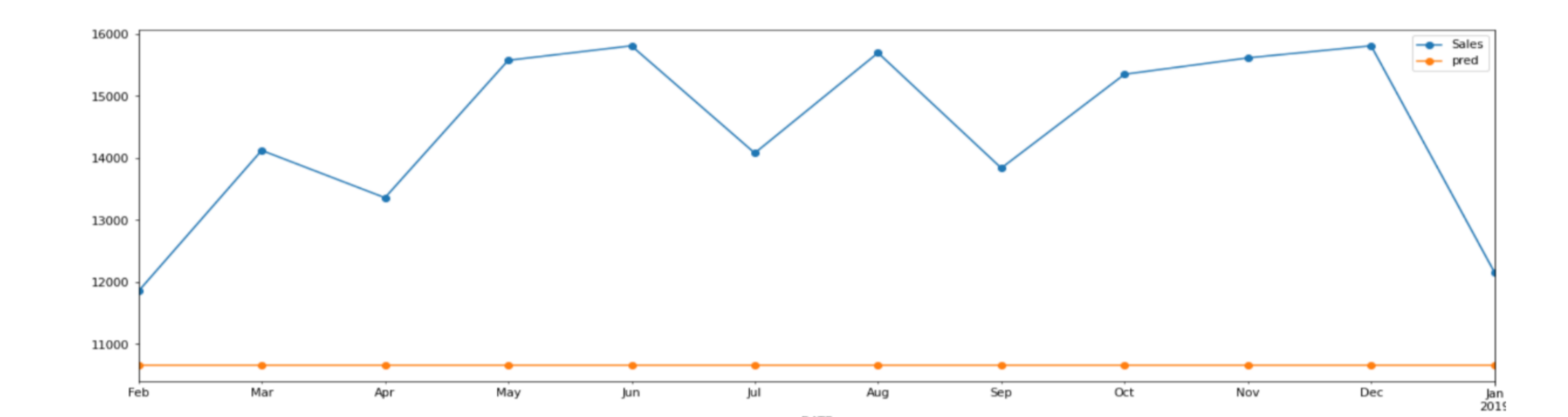
In **multiplicative seasonality,** we obtain a time series in which the amplitude of the **seasonal component is increasing with an increasing trend.**

5.2 Simple Methods for Forecasting:

Naive — The forecasts are equal to the **last observed data.**



Mean/Median — The forecasts are equal to the **mean/median of observed data.**



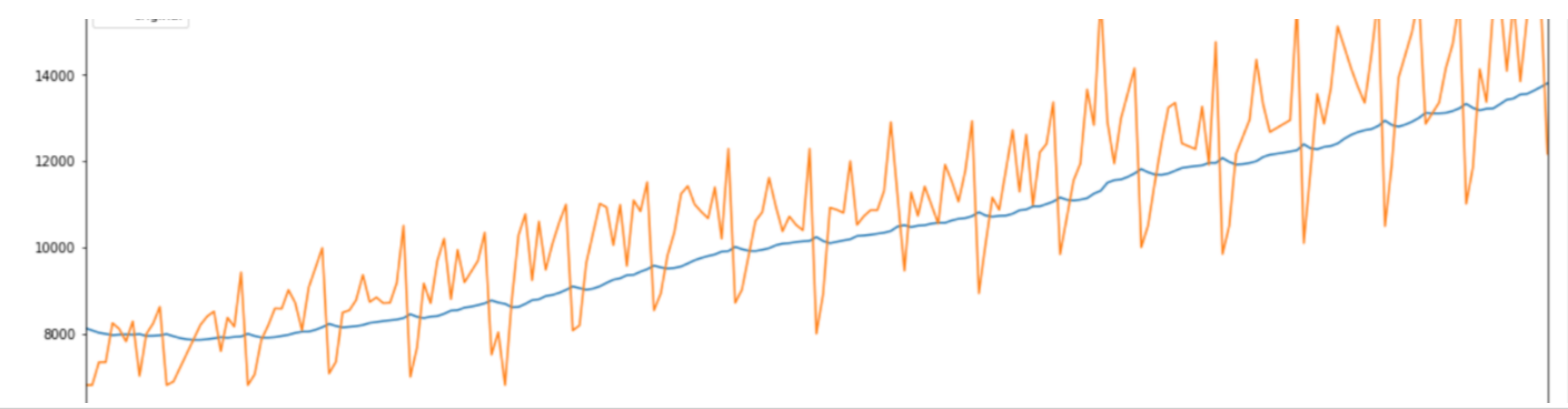
Season Naive — The forecasts are **equal to the observed value** at **the same time** from the last occurrence of same season.

5.3 Effective Forecasting Methods (Exponential Smoothing):

Simple Exponential Smoothing

The key idea is to not only keep some memory of the entire time series, but also to give **more value** to the **recent data** and **less value** to the **past value.**

$$\hat{y}_{t+h} = \alpha y_t + \alpha(1-\alpha)y_{t-1} + \alpha(1-\alpha)^2\hat{y}_{t-2} + \dots$$



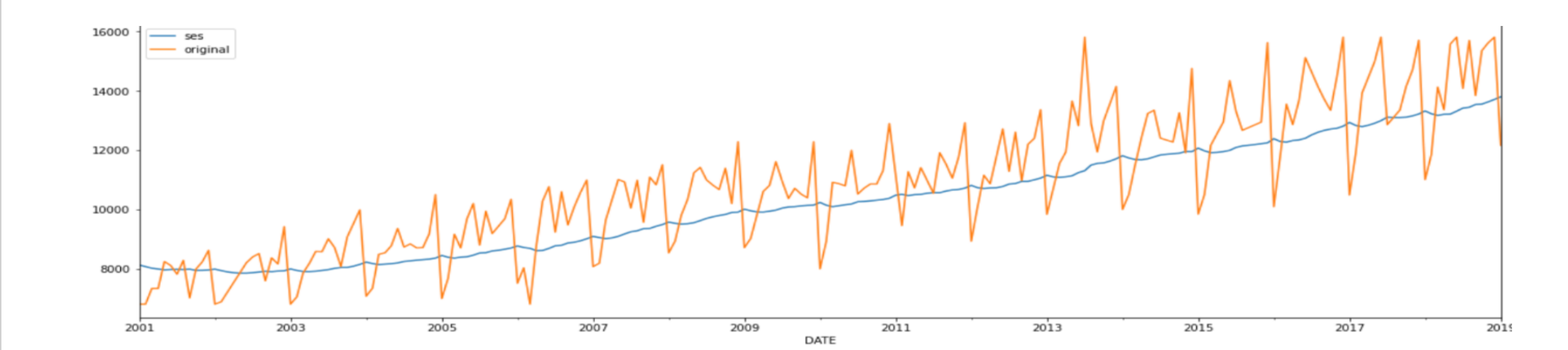
Double Exponential Smoothing

The trend of the entire time series in the **SES formulation** is incorporated to forecast future values.

$$\hat{y}_{t+h} = l_t + hb_t$$

where

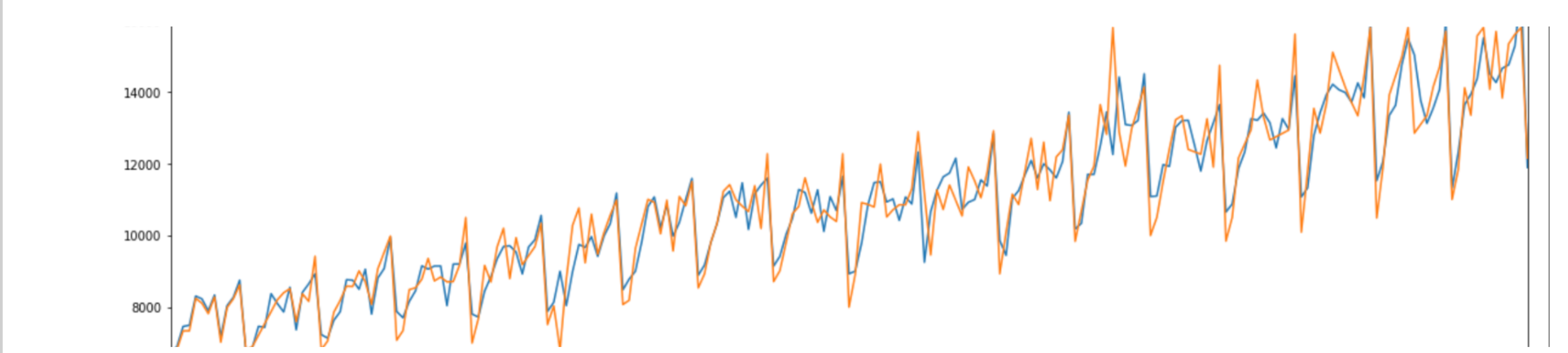
$$l_t = \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1})$$



Triple Exponential Smoothing

Triple Exponential Smoothing is an extension of Double Exponential Smoothing that explicitly **adds support for seasonality** to the univariate time series.

$$\hat{y}_{t+h} = l_t + hb_t + s_{t+h-m}$$



5. 4 Stationarity

- A time series whose properties are **not dependent upon time**.
- Therefore time series with trend and seasonality are non stationary.
- Differencing can be used to remove non stationarity. In first differences the values become the difference between consecutive original values.
- Similarly second differences we find the difference between the consecutive values of first differences.

5.5 ARIMA Forecasting methods:

AR (Autoregressive model)

In AR models, the variable of interest is forecasted using a **linear combination of past value** of the variable

$$y_t = c + \varnothing_1 y_{t-1} + \varnothing_2 y_{t-2} + \ldots + \varnothing_p y_{t-p} + \varepsilon_t$$

The above equation shows the AR model of order p, i.e. AR(p)

MA (Moving Average)

In MA models, we use the **past forecast errors** for forecasting.

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \ldots + \theta_q \varepsilon_{t-q}$$

In MA models, we use the past forecast errors for forecasting.

ARMA (Auto Regressive Moving Average)

It is used to describe **stationary time series** in terms of AR and MA.

- In ARMA (p, q), p is the order of AR and q is the order of MA.
- It includes lagged values as well as the lagged errors.

$$X_t = c + \varepsilon_t + \sum_{i=1}^{p} \varphi_i X_{t-i} + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i}$$

ARIMA (Auto Regressive Integrated Moving Average)

ARIMA is combination of AR and MA alongwith integration which is opposite of differencing.

$$y'_t = c + \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

Here the predictors include both the lagged values of y and lagged errors. Here is the differenced series.
In **ARIMA** (**p, d, q**) means that:

- **p** is the **order of AR model**
- **d** is the **degree first differencing**
- **q** is **order of MA model**

Its different from ARMA in the aspect that ARMA requires the time series to be stationary.

SARIMA (Seasonal Auto regressive integrated moving average):

SARIMA model can model **seasonal data**. Its formed by adding seasonal terms in the ARIMA model.

$$y'_t = c + \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

SARIMA can be represented by,

- where m = seasonal period
- upper case notations are for seasonal terms
- lower case notations are for non-seasonal terms

The seasonal part involves terms similar to non-seasonal terms but involves backshift of the seasonal period.

ARIMAX (ARIMA + Exogenous variable)

Exogenous variables are variables whose cause is external to the model and whose **role is to explain other variables** or outcomes in the model

$$y_i = \beta x_i + \sum_{j=1}^{p} \phi_j y_{i-j} + \varepsilon_i + \sum_{j=1}^{q} \theta_j \varepsilon_{i-j}$$

Here,

- x is an exogenous variable used along with lagging errors and lagging values.
- There is also SARIMAX (SARIMA + Exogenous variable